

# ABSTRACT

## **Title: Smart Industrial Monitoring System with Predictive Maintenance Using CAN FD Network and Edge Machine Learning Analytics**

The industrial sector faces significant challenges due to unexpected equipment failures causing costly downtime and inefficient maintenance practices. This project presents a smart industrial monitoring system that leverages high-speed CAN FD networking for real-time sensor data collection from temperature, vibration, and power sensors on industrial equipment. The collected data is transmitted to a central Raspberry Pi equipped with edge machine learning models for anomaly detection and predictive maintenance scheduling. Key machine learning algorithms including Random Forest and LSTM are employed to identify early fault signs and reduce unplanned downtime. The system features a cloud-connected dashboard providing real-time visualization and instant alerts to stakeholders, enhancing decision-making. The proposed approach aims to improve equipment uptime by 30%, reduce maintenance costs by 30%, and increase overall operational efficiency. This solution effectively combines robust industrial communication protocols with intelligent edge analytics to transform traditional maintenance methods into proactive, data-driven operations.

Keywords: CAN FD, Industrial IoT, Predictive Maintenance, Edge Computing, Machine Learning, Anomaly Detection, Real-time Monitoring, Industrial Automation, Raspberry Pi, Sensor Networks

**Submitted By: (Project Group No: A26)**

|    | <b>Roll No.</b> | <b>Name</b>                  |
|----|-----------------|------------------------------|
| 1. | E41100          | Chandrakant Janardhan Shinde |
| 2. | E41103          | Aditya Shivhari Deoche       |
| 3. | E41104          | Umesh Shivaji Bhabad         |

|                                                  |                                                           |                                                              |
|--------------------------------------------------|-----------------------------------------------------------|--------------------------------------------------------------|
| Mrs. B. A. Patil                                 | Mrs. S. R. Jadhav<br>Mr. S. K. Shinde<br>Mr. P. S. Kokare | Dr. Achala Deshmukh                                          |
| <b>Project Guide</b><br><b>Dept. of E&amp;TC</b> | <b>Project Coordinator</b><br><b>Dept. of E&amp;TC</b>    | <b>Program Coordinator (HOD)</b><br><b>Dept. of E&amp;TC</b> |